PDF Wizard

Document Workflows in the Browser

Technical Overview and Product Report

Prepared by the Document Platform Team

January 2026 | Version 1.0

Abstract

PDF Wizard brings the core capabilities of a desktop PDF editor to any modern browser. This report explains how documents are opened, rendered, annotated, filled, signed and exported without installing software, and how a small companion server handles the few tasks that browsers cannot perform well, such as converting office files, recognizing text in scans and compressing large documents. We describe the architecture, walk through common review and form workflows, summarize performance measurements and outline the work that remains.

For internal distribution. Sample document generated for automated tests.

Contents

1 Introduction **3**

1.1 Background 3

1.2 Goals 3

2 Document Workflows **4**

2.1 Reviewing and Commenting 4

2.2 Preparing Forms 4

3 Performance **5**

3.1 Benchmarks 5

4 Conclusion **6**

4.1 Further Reading 6

About this report

This report is intended for product managers, engineers and administrators who are evaluating PDF Wizard Web for their organizations. Every entry in the list above is a link: select an entry to jump to the corresponding page. The bookmarks panel shows the same structure and can be expanded to reveal subsections.

1 Introduction

Portable Document Format files remain the common currency of business, government and education. Contracts, invoices, research papers and application forms are exchanged as PDF because the format preserves layout exactly, embeds fonts and images, and can be displayed identically on every device. Yet the tools used to work with these files have traditionally been desktop applications that must be installed, licensed and updated on each machine.

PDF Wizard takes a different approach. The entire editor runs inside the browser tab, using a JavaScript rendering engine for display and a separate library for writing changes back into the file. Documents never leave the device unless the user explicitly saves them to the shared library or requests a server-side conversion.

1.1 Background

Early browser viewers could display a PDF but offered little beyond scrolling and printing. Over the last decade, rendering engines have matured to the point where text selection, search, form filling and accurate font rendering are available in every major browser. At the same time, teams have moved their work into web applications, and switching to a separate desktop program just to comment on a document has become a noticeable interruption.

Several needs appear again and again in interviews with users: reviewing a draft with colleagues, filling and signing a form, combining several files into one package, and removing sensitive information before a document is shared outside the organization.

1.2 Goals

The project set out to deliver the familiar PDF Wizard experience with three guiding goals. First, fidelity: pages must look exactly as they do in the desktop product, including rotated pages, unusual page sizes and embedded fonts. Second, responsiveness: opening a large report should feel instant, and editing tools should react within a single animation frame. Third, safety: every change must be reversible until the user saves, and the original file must never be altered in place.

2 Document Workflows

A workflow is the sequence of steps a document passes through between its first draft and its final, approved form. PDF Wizard supports the workflows that users mentioned most often, and each one is available from the All tools page as well as from the quick tools rail beside the document.

2.1 Reviewing and Commenting

Reviewers can highlight, underline or strike out text, attach sticky notes, draw shapes and add free-form ink. Every comment records its author and the time it was made, and replies are threaded under the original note. A typical review follows these steps:

1. Open the draft from the library or drag it into the browser window.
2. Select the Comment tool and highlight passages that need attention.
3. Add sticky notes with questions, and reply to notes left by other reviewers.
4. Filter the comment list by author or status to keep track of open issues.
5. Save the reviewed copy, or export the comments as a summary for the author.

Because comments are stored as standard PDF annotations, a document reviewed in PDF Wizard can be opened in any other conforming viewer and every note will appear in the expected place.

2.2 Preparing Forms

Many organizations still distribute paper forms that have been scanned into PDF. The form preparation tool detects labels and boxes on such pages and proposes text fields, check boxes and signature areas. Authors can then adjust each field, set its name and tooltip, and mark it as required. Recipients fill the form directly in the browser, and the values are written into the file when it is saved.

3 Performance

Performance was measured on a mid-range laptop using a set of representative documents, ranging from a two-page letter to a three-hundred-page technical manual. Each operation was repeated twenty times after a warm-up run, and the median and 95th percentile durations were recorded. Table 1 summarizes the results.

3.1 Benchmarks

Operation	Document	Median (ms)	95th pct (ms)
Open and render first page	300 pages	180	240
Render page at 200% zoom	6 pages	45	70
Full-text search	300 pages	310	420
Add 50 comments	6 pages	22	35
Export with annotations	6 pages	140	190
Compress (server)	42 MB	2,900	3,600

Table 1: Median and 95th percentile timings for common operations.

Rendering dominates the cost of opening a document, so pages are drawn lazily as they scroll into view and thumbnails are generated in the background at low resolution. Text for search is extracted page by page and cached, which keeps the first search in a long document under half a second. Operations that rewrite the file, such as exporting or compressing, run away from the main thread so that scrolling stays smooth.

The largest remaining cost is server-side compression of image-heavy scans, which is bounded by the time needed to re-encode each image. PDF Wizard shows progress for such tasks and lets the user continue working while they complete.

4 Conclusion

PDF Wizard demonstrates that a complete document editor can live comfortably in the browser. Viewing, commenting, form filling, page organization and export all run locally, while a small server provides conversion, text recognition and compression when they are needed. Early users report that the tools feel familiar and that removing the installation step has made it easier to involve occasional reviewers.

Future work will focus on accessibility tagging, side-by-side comparison of document versions, and deeper integration with electronic signature services. We also plan to publish the benchmark documents so that others can reproduce the measurements in Table 1.

4.1 Further Reading

The PDF specification is maintained as the international standard ISO 32000. Additional material, including release notes and a list of supported features, is available on the project website:

> https://example.com

Questions and feedback about this report can be sent to the Document Platform Team. Thank you for reading.